

Validation Architecture in delaunay

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The validation architecture in `delaunay` separates local object validity, combinatorial consistency, intrinsic piecewise-linear topology, embedding validity, and geometric predicates. This separation is useful both mathematically and in software: PL-manifoldness remains an intrinsic property of the abstract simplicial complex, while Euclidean, toroidal, spherical, and future embedding models specialize only the extrinsic realization and geometric predicate layers.

Additional Key Words and Phrases: Delaunay triangulation, validation, piecewise-linear topology, computational geometry, Rust

1 VALIDATION ARCHITECTURE

The central design principle is that validation layers answer different questions. The first three layers are independent of the embedding, while the last two are embedding-dependent. This decomposition lets the same triangulation data structure and bistellar move machinery serve Euclidean, toroidal, and spherical workflows without redefining PL topology for each ambient model.

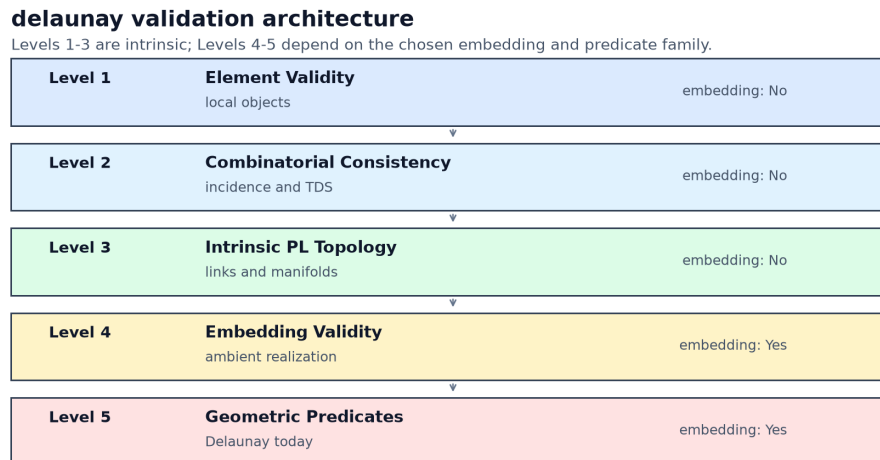


Fig. 1. The five validation levels in `delaunay`. Levels 1–3 are intrinsic to the finite simplicial complex; Levels 4–5 depend on the selected embedding and geometric predicate family.

1.1 Element Validity

Element Validity asks whether individual objects are internally well-formed. For vertices this includes finite coordinates, a non-nil UUID, and the expected dimension. For simplices this includes exactly $d + 1$ distinct vertex references, valid local identity, and any owner-local orientation convention. This layer is deliberately local: it does not decide

whether two simplices fit together, whether the complex is a manifold, or whether the coordinates realize the complex without overlap.

1.2 Combinatorial Consistency

Combinatorial Consistency asks whether the triangulation data structure represents a coherent abstract simplicial complex. This includes key/UUID mappings, simplex incidence, reciprocal neighbor references, bounded facet multiplicity, ridge connectivity, and coherent combinatorial orientation. These checks still do not depend on a metric or embedding. A complex can pass this layer and still fail to be a PL manifold or fail to embed faithfully.

1.3 Intrinsic PL Topology

Intrinsic PL Topology asks whether the abstract complex has the requested PL topological type. For a k -simplex σ in a d -dimensional PL-manifold with boundary, the link condition is

$$\text{lk}(\sigma) \cong \begin{cases} S^{d-k-1}, & \sigma \text{ is interior,} \\ B^{d-k-1}, & \sigma \text{ is boundary.} \end{cases}$$

The implementation enforces this through facet-degree checks, boundary closure, ridge-link and vertex-link certification as configured by `TopologyGuarantee`, connected components, isolated-vertex rejection, and Euler characteristic checks. These invariants are the PL-topology layer of the library, in the same spirit as the intrinsic bistellar-equivalence theory underlying Pachner moves [4, 5].

This layer is intrinsic. It is unchanged by replacing a Euclidean embedding with a toroidal or spherical one. In particular, spherical support does not require a new definition of PL-manifoldness; it requires spherical embedding and spherical Delaunay predicates in the later layers.

1.4 Embedding Validity

Embedding Validity asks whether the abstract complex is faithfully realized in the selected ambient model. In Euclidean affine charts this requires nondegenerate maximal simplices and no intersections except along shared faces. Toroidal embeddings apply the same idea in periodic covering-space charts. Spherical embeddings instead require vertices to lie on $S^d \subset \mathbb{R}^{d+1}$ and require spherical cells to be nondegenerate on that embedded sphere.

This layer is separate from Delaunay-ness. A triangulation can be a valid embedding without being Delaunay, and a Delaunay predicate is not meaningful on a folded or self-intersecting embedding.

1.5 Geometric Predicates

Geometric Predicates ask whether a valid embedding satisfies a selected geometry-specific condition. The implemented predicate family today is Delaunay. In Euclidean and toroidal affine charts this is the empty circumsphere condition, implemented with robust geometric predicates in the style of Shewchuk [6] and classical Delaunay triangulation theory. Intrinsic Delaunay constructions on Riemannian manifolds provide the broader geometric setting for this separation [3]. For spherical d -D Delaunay triangulations, input points are normalized in $\mathbb{R}^{d+1}$, the construction uses the ambient convex hull, and spherical cells correspond to outward hull facets [1, 2]. The intrinsic dimension remains d ; the extra coordinate belongs to the spherical embedding backend, not to the abstract simplex dimension.

Future predicate families, such as regular triangulations, weighted Delaunay, Gabriel graphs, alpha complexes, constrained Delaunay variants, or hyperbolic predicates, should specialize this layer without changing the first three layers.

1.6 Diagnostics and Failure Examples

The developer-facing API exposes layer-local fast-fail checks and aggregate reports. Figure 2 is generated by the validation notebook from the same command-line artifact used in tests, so the figure tracks the public diagnostics instead of hand-drawn examples.

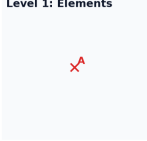
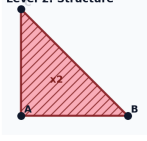
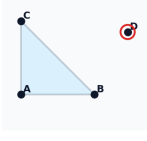
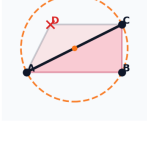
Generated failure picture	Public check / test	Explanation
Level 1: Elements 	<code>Point::<2>::try_new src/geometry/point.rs::point_is_valid_f64</code>	Non-finite point coordinate. Element validation rejects non-finite coordinates before they can enter a vertex, simplex, or TDS. Diagnostic: Invalid coordinate at index 0 in dimension 2: NaN
Level 2: Structure 	<code>DelaunayTriangulationBuilder::try_from_vertices_and_simplices(...) build_tests/triangulation_builder.rs::test_explicit_error_variant_duplicate_simplices_structural_validation</code>	Duplicate maximal simplex. The TDS layer rejects duplicate maximal simplices because the incidence structure would no longer be a well-defined complex. Diagnostic: TDS assembly failed during explicit construction. Validation error during construction: Duplicate simplices detected: Refusing to insert duplicate simplex <uuid> with same vertex UUIDs as existing simplex SimplexKey(1v1): ([<uuid>, [0, 0]], [<uuid>, [0, 0]], [<uuid>, [0, 0]])
Level 3: Topology 	<code>Triangulation::is_valid_topology tests/triangulation_builder.rs::test_explicit_unreferenced_vertices_rejected</code>	Unreferenced vertex. The topology layer rejects isolated vertices because every vertex must belong to the triangulated space. Diagnostic: Topology validation failed during explicit construction: Isolated vertex: vertex <uuid> (key VertexKey(4v1)) is not incident to any simplex
Level 4: Valid affine realization 	<code>Triangulation::validate_embedding tests/triangulation_builder.rs::test_explicit_error_variant_geometric_nondegeneracy</code>	Degenerate realized simplex. The coordinate realization is not a valid affine realization because the abstract 2-simplex collapses to zero area. Diagnostic: Geometric nondegeneracy validation failed during explicit construction: Degenerate geometric orientation: Simplex <uuid> (key SimplexKey(1v1)) is geometrically degenerate (zero-volume simplex from collinear/coplanar vertices)
Level 5: Delaunay 	<code>DelaunayTriangulation::is_valid_delaunay tests/triangulation_builder.rs::test_explicit_non_delaunay_mesh</code>	Interior point in a circumcircle. Point D lies inside the circumcircle of triangle ABC, so the chosen diagonal violates the local Delaunay property. Diagnostic: Delaunay validation failed during explicit construction: Delaunay verification failed: empty-circumsphere validation failed: Simplex SimplexKey(1v1) violates Delaunay property; offending vertex: Some(VertexKey(4v1))

Fig. 2. Generated validation failures by layer, refreshed from the validation notebook and command-line diagnostic artifact.

1.7 Extensibility

The decomposition can be summarized as

$$\text{PL topology} \perp \text{embedding topology} \perp \text{geometric predicates.}$$

This is the useful software boundary. Bistellar moves operate on the abstract simplicial complex and preserve PL topology when their local preconditions are met. Embedding backends then decide whether the edited complex is faithfully realized, and predicate backends decide whether the realized complex is Delaunay or satisfies another geometric condition.

ACKNOWLEDGMENTS

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REFERENCES

- [1] Manuel Caroli, Pedro M. M. de Castro, Sylvain Pion, and Monique Teillaud. 2010. Robust and Efficient Delaunay Triangulations of Points on or Close to a Sphere. *International Journal of Computational Geometry & Applications* 20, 4 (2010), 329–356.
- [2] Herbert Edelsbrunner. 2001. *Geometry and Topology for Mesh Generation*. Cambridge University Press, Cambridge, United Kingdom.
- [3] Greg Leibon and David Letscher. 2000. Delaunay triangulations and Voronoi diagrams for Riemannian manifolds. In *Proceedings of the sixteenth annual symposium on Computational geometry*. ACM, Clear Water Bay Kowloon Hong Kong, 341–349. <https://doi.org/10.1145/336154.336221>
- [4] W. B. R. Lickorish. 1999. Simplicial moves on complexes and manifolds. <https://doi.org/10.48550/arXiv.math/9911256> arXiv:math/9911256.
- [5] Udo Pachner. 1991. P.L. Homeomorphic Manifolds Are Equivalent by Elementary Shellings. *European Journal of Combinatorics* 12, 2 (1991), 129–145.
- [6] Jonathan Richard Shewchuk. 1997. Adaptive Precision Floating-Point Arithmetic and Fast Robust Geometric Predicates. *Discrete & Computational Geometry* 18, 3 (1997), 305–363.